

Perfect Circle — Unit Test Document

Prepared by: Manya Bansal · Sep 27, 2026 · Version 0.1

48 unit tests cover the scoring, validation, modes, daily challenge, progression, duel logic, data and local server modules; they run with Vitest on every push and must all pass before a deploy.

Strategy

- Tool: Vitest with v8 coverage. Run locally with `npx vitest run --coverage`.
- Test strokes are generated by helper functions (perfect circle, noisy circle, square, star) with a fixed random seed, so results are the same on every run.
- Time-based logic (timers, fading ink, disconnects) uses Vitest fake timers. IndexedDB uses the fake-indexeddb library. Server routes are tested with Supertest against an in-memory SQLite database; duel rooms are tested with two local Socket.IO test clients.
- Canvas drawing and screens are not unit tested; Playwright end-to-end tests cover them.

Coverage targets

Module	Line coverage target
scoring	95%
daily, progression	90%
input, modes, duel, data, server	80%
Whole project	80%

Entry and exit criteria

- Entry: the module compiles with no TypeScript errors, and its functions match the Modules table in the technical design.
- Exit: all 48 tests pass, coverage targets are met, and no open defect is rated Critical or High.

Test cases

ID	Module	Test case	Expected result
UT-01	input	Resample 37 unevenly spaced points	128 points; spacing equal within 1%
UT-02	input	Resample a stroke of 1 point	Returns empty stroke, no crash
UT-03	input	Capture 2 points less than 2 px apart	Second point ignored
UT-04	scoring	Perfect generated circle, radius 150 px	Score 99.9% or more
UT-05	scoring	Circle with uniform noise of plus or minus 5% of radius	Score between 92% and 95%
UT-06	scoring	Ellipse 150 x 100 px scored as a circle	Score below 75%
UT-07	scoring	50 random scribbles	Every score between 0 and 100
UT-08	scoring	Same circle at radius 80 px and 300 px	Scores equal within 0.1
UT-09	scoring	Perfect square	Score 99% or more
UT-10	scoring	Perfect square rotated 37 degrees	Score 99% or more (rotation fitted)
UT-11	scoring	Circle drawn for a square target	Corner coverage below 1; score below 75%
UT-12	scoring	Perfect triangle	Score 99% or more
UT-13	scoring	Perfect 5-point star	Score 99% or more
UT-14	scoring	Star with one arm missing	Corner coverage 0.8 applied to score
UT-15	scoring	Stroke with a point 30 px from the dot	Rejected: too_close

ID	Module	Test case	Expected result
UT-16	scoring	Stroke going back 0.50 rad	Rejected: wrong_way
UT-17	scoring	Stroke going back 0.40 rad	Accepted (boundary)
UT-18	scoring	Stroke covering 90% of a turn	Rejected: not_closed
UT-19	scoring	Stroke covering exactly 93% of a turn	Accepted (boundary)
UT-20	scoring	Classic attempt of 10.1 s; timed 3 s attempt of 3.01 s	Both rejected: too_slow
UT-21	scoring	Stroke of 9 points	Rejected: too_short
UT-22	modes	inkAlpha at 0 s, 0.2 s, 0.4 s	1, 0.5, 0
UT-23	modes	dotPosition over 10 s, called twice	Same positions both times; speed 30 px/s within 5%
UT-24	modes	Perfect circle drawn around the moving dot's live position	Score 99% or more
UT-25	modes	3 s timer with fake timers	Expires at 3000 ms, not before
UT-26	daily	dailyChallenge for 2026-10-01 called 3 times	Same shape and mode each time
UT-27	daily	dailyChallenge for 365 days	All 4 shapes and all modes appear
UT-28	daily	23:59:59 UTC vs 00:00:00 UTC next day (05:29:59 vs 05:30:00 IST)	Different challenges
UT-29	daily	shareText for challenge 42, score 91.84	Contains "#42", "91.8%" and exactly 5 squares
UT-30	daily	Segment errors of 2%, 5%, 8%	Green, yellow, red
UT-31	progression	Dailies on 3 consecutive days	Streak 3

ID	Module	Test case	Expected result
UT-32	progression	Streak 5, then a missed day, then a daily	Streak 1; best streak stays 5
UT-33	progression	Two dailies on the same UTC day	Streak goes up by 1 only
UT-34	progression	Scores 89.9% and 90.0%	Sharp eye badge only for 90.0%
UT-35	progression	Two scores above 95%	Nearly perfect awarded once
UT-36	progression	85%+ on 3 shapes, 84% on the 4th	Shape master not awarded
UT-37	progression	historySeries on a 10-day fixture	Daily best and 7-day average match hand-calculated values
UT-38	progression	historySeries with off-hand filter on	Only off_hand attempts returned
UT-39	duel	Generate 1,000 room codes	4 characters each; no O, 0, l or 1
UT-40	duel	assignTouch at x = 100 and x = 900 on a 1000 px screen; touch crossing the middle	Players 1 and 2; a touch keeps its starting side
UT-41	duel	Round results win, win; and a tied round	Match ends after 2 wins; tie goes to faster finish
UT-42	duel	Opponent disconnected for 15.1 s (fake timers)	Remaining player wins
UT-43	data	saveAttempt with fake IndexedDB	Attempt stored and added to sync queue
UT-44	data	syncPending with network success, then failure	Queue cleared on success; kept on failure
UT-45	data	POST /api/daily with the real stroke, then with a changed client score	Server score matches within 0.01; changed score ignored
UT-46	server	POST /api/login with a new nickname, then the same nickname with a wrong PIN	Player created with a hashed PIN; wrong PIN rejected with status 401

ID	Module	Test case	Expected result
UT-47	server	Two Socket.IO test clients join the same room code	Both receive the same countdown start time
UT-48	server	POST /api/attempts twice with the same client_ids	Each attempt stored once

Sample test

```
import { describe, it, expect } from 'vitest';
import { scoreCircle } from '../src/scoring/circle';
import { makeCircle } from '../helpers/strokes';

describe('scoreCircle', () => {
  it('UT-04 scores a perfect circle at 99.9% or more', () => {
    const stroke = makeCircle({ cx: 400, cy: 400, r: 150, points: 128 });
    expect(scoreCircle(stroke, { cx: 400, cy: 400
  })).toBeGreaterThanOrEqual(99.9);
  });
});
```

Defect logging

Failed tests are logged as GitHub Issues with the test ID, input, expected and actual result, and a severity of Critical, High, Medium or Low.